



Salifort Motors: Employee Attrition Analytics

Workforce Trends, Attrition Drivers & Retention Insights



Salary

Low

Medium

High

Department

All

Promotion Status

All



16.6%

Attrition Rate



0.63

Avg Satisfaction



12K

Total Workforce



2K

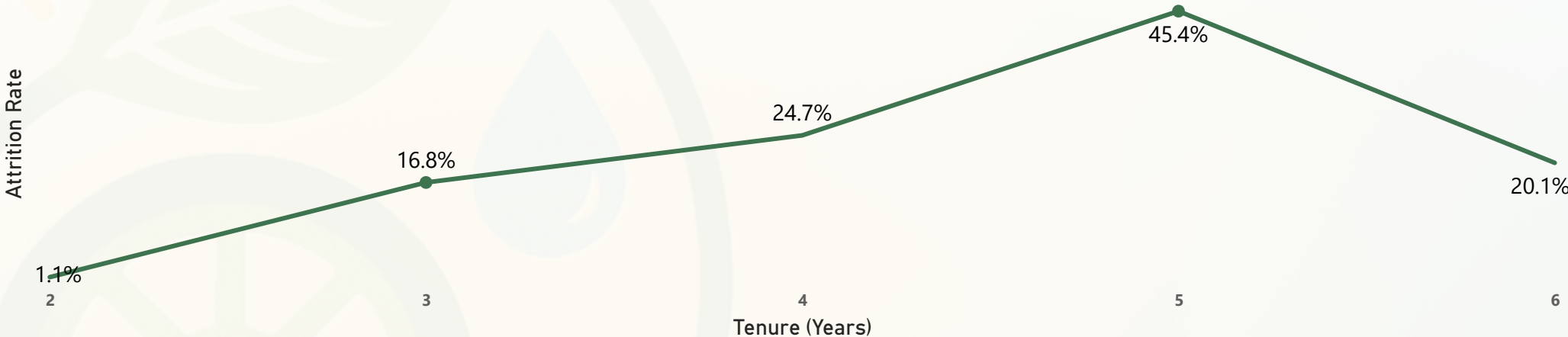
Employees Lost



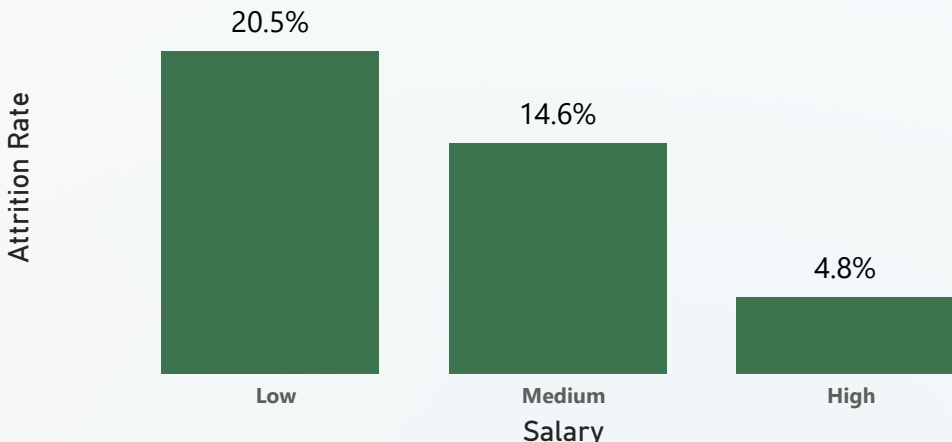
200.47

Avg Monthly Hrs

Attrition Rate by Tenure (Years)



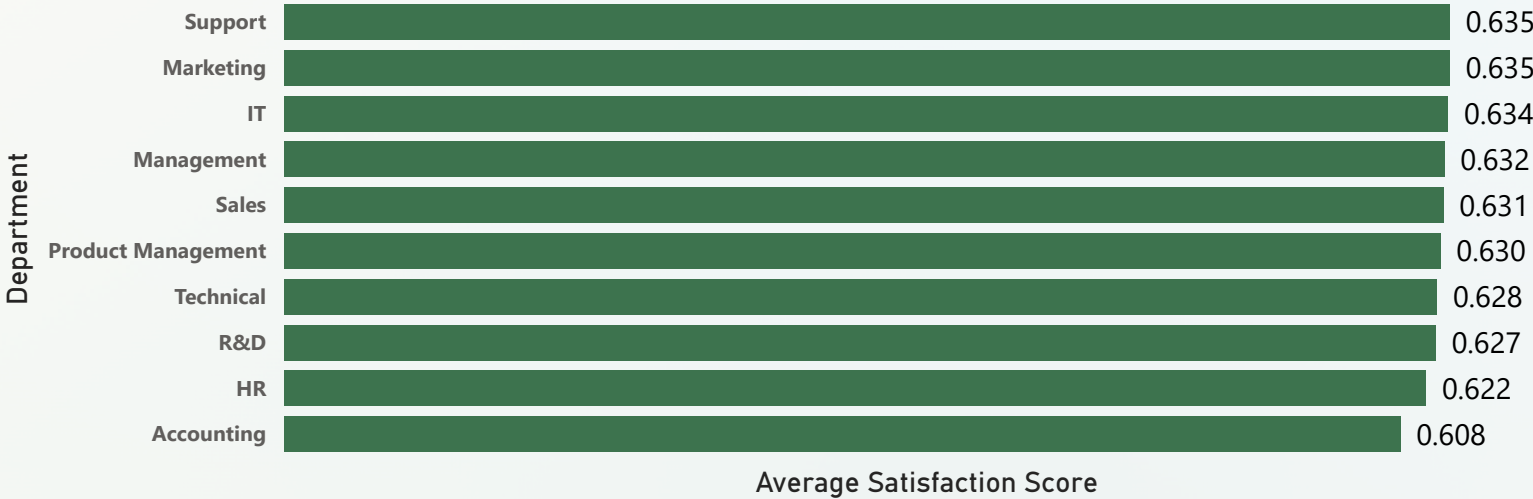
Attrition Rate by Salary Level



Attrition Rate by Department



Average Satisfaction Score by Department





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Understanding the Operational Factors Behind Employee Attrition



Salary

Low

Medium

High

Department

All

Tenure

All



8K
Overworked



988
Top Talent Lost



3.8
Avg Project Load

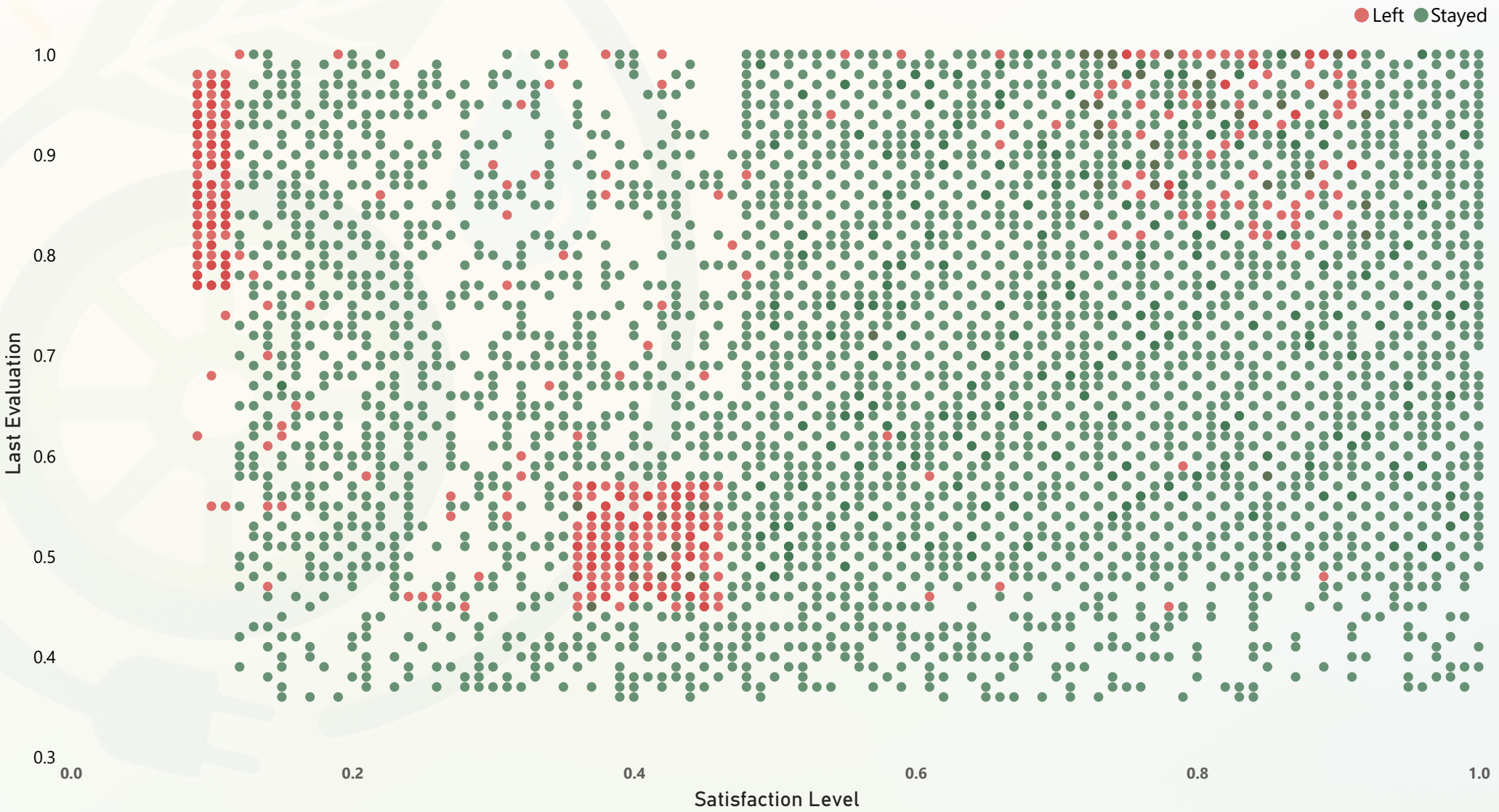


1.69%
Promotion Rate

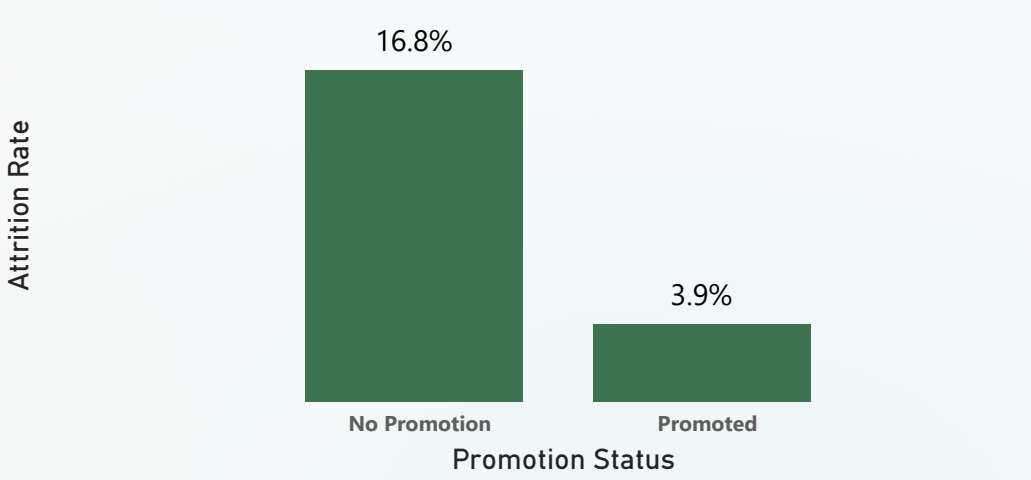


0.72
Avg Eval Score

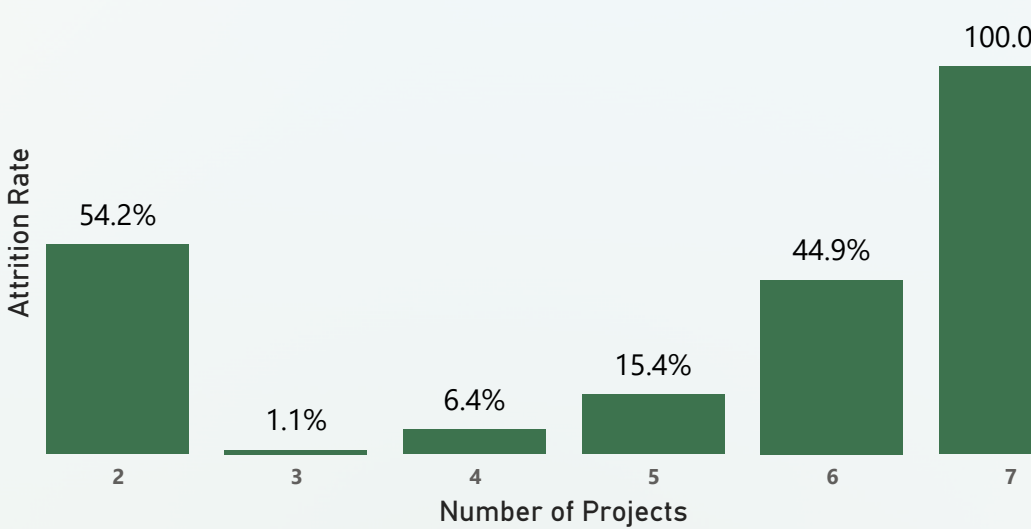
Employee Satisfaction vs Performance Evaluation



Attrition Rate by Promotion Status



Attrition Rate by Number of Projects





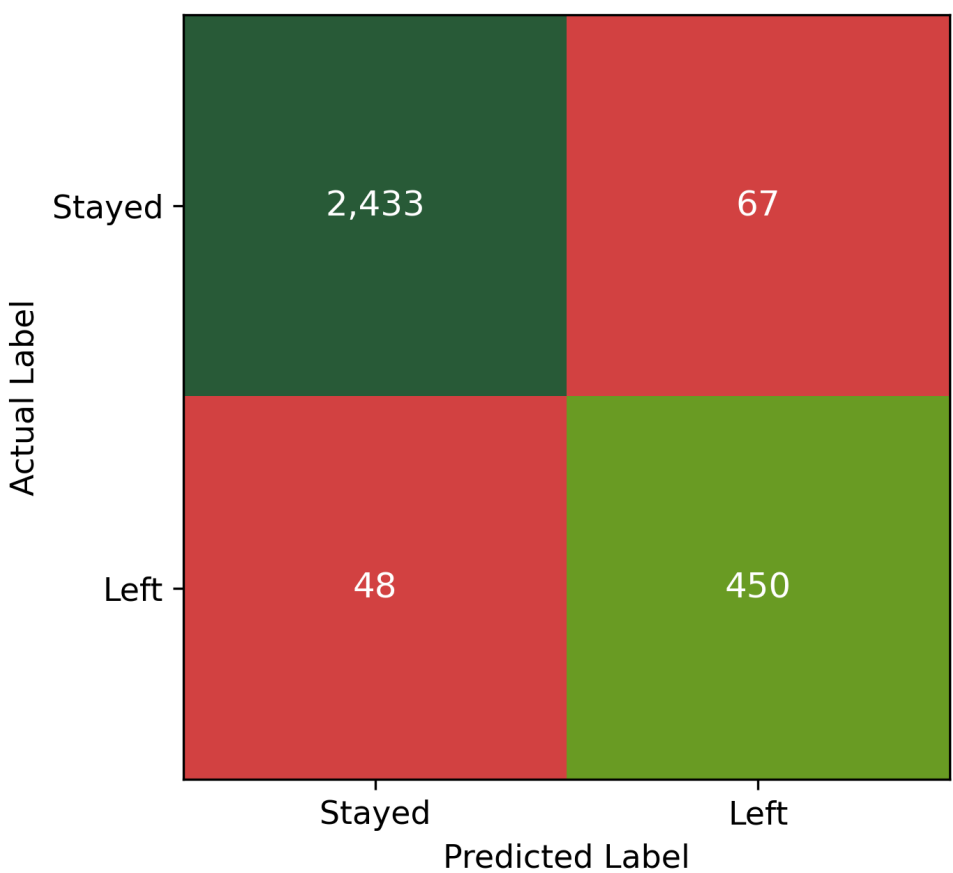
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Identifying Employees at Risk of Leaving Using Machine Learning

Model Performance Comparison

Model Name	Recall	Precision	F1 Score	Accuracy
Random Forest (Champion)	90.36%	87.04%	88.67%	96.16%
Decision Tree	91.77%	78.39%	84.55%	94.43%
Logistic Regression	26.00%	44.00%	33.00%	82.00%

Random Forest Confusion Matrix



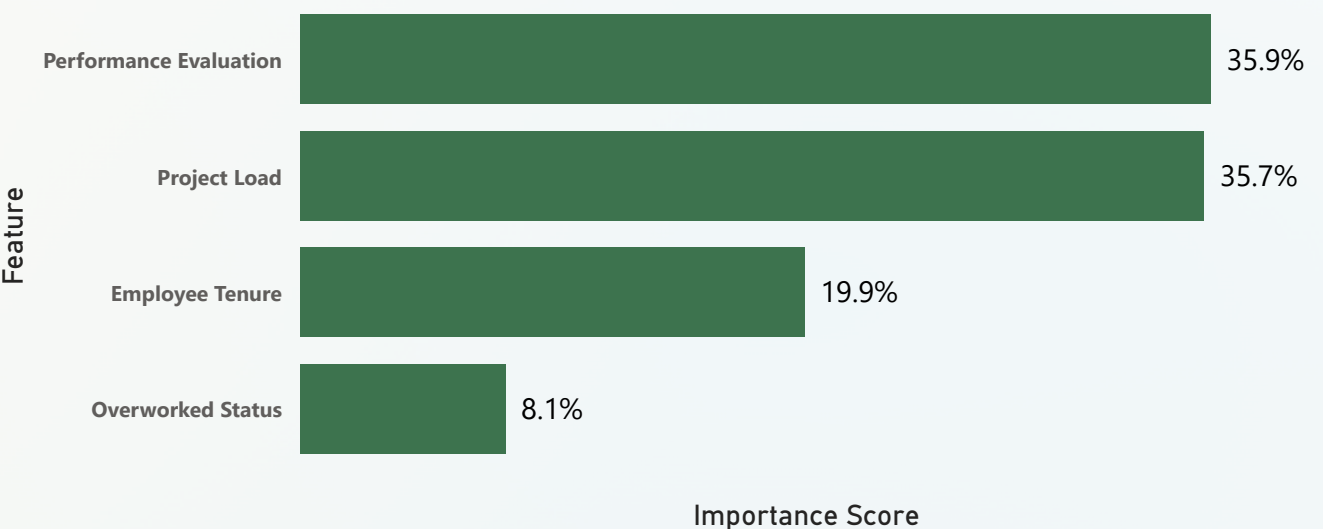
Business Impact of the Model

- **High Detection Rate:** Successfully identified **450 (90%) of employees who ultimately left**, missing only **48 attrition cases**.
- **Strategic Trade-off:** Prioritizes catching costly flight risks while keeping false alarms (**67 false positives**) at a highly manageable level.
- **Resource Optimization:** Correctly identified **2,433 stable employees**, ensuring that retention budgets and HR efforts are focused precisely where they are needed.
- **HR Enablement:** Empowers leadership to shift away from reactive backfilling toward proactive retention strategies.

Model Selection Insight

Random Forest was selected as the **champion model** due to its strong balance of **Recall (90.36%)**, **Precision (87.04%)**, and **F1 Score (88.67%)**. While the Decision Tree achieved slightly higher Recall, Random Forest delivered better overall predictive performance and a **lower false-positive rate**, making it the preferred model for employee retention strategies.

Feature Importance Drivers



Analytical Conclusion & Readiness

The machine learning phase validates that employee turnover at Salifort Motors is a **predictable operational challenge** rather than an unpredictable human factor. With **96.16% Accuracy**, **90.36% Recall**, and **87.04% Precision**, the **Champion Random Forest model** serves as a reliable early-warning system for attrition. By identifying **performance evaluation**, **project load**, **employee tenure**, and **overworked status** as the primary drivers of turnover, the analysis provides a foundation for **targeted retention strategies**.

Feature Importance Insight

Performance Evaluation (35.9%) and **Project Load (35.7%)** emerged as the strongest attrition drivers, followed by **Employee Tenure (19.9%)** and **Overworked Status (8.1%)**. These findings indicate that **workload intensity**, **performance pressure**, and **tenure-related factors** have a greater influence on employee turnover than compensation or department-level differences.



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Translating Analytical Insights into Actionable Retention Strategies

Strategic Retention Recommendations

1. Implement Workload Caps

Limit assignments to a maximum of 4 projects per employee. Monitor workloads exceeding 175 monthly hours and establish sustainable overtime policies to reduce burnout risk.

2. Mid-Tenure Career Pathways

Develop targeted promotion, mentorship, and career-development programs for employees within the critical 4–6 year tenure window.

3. Recognition & Overtime Reform

Reward quality of outcomes rather than hours worked. Ensure high-performing employees are not consistently burdened with unsustainable workloads.

4. Holistic Performance Reviews

Incorporate workload, engagement, and burnout indicators into performance discussions to identify flight risks before turnover occurs.

5. Predictive HR Monitoring

Deploy the Random Forest model as an early-warning system to proactively identify high-risk employees and enable timely retention interventions.

The recommended initiatives directly address the key drivers identified through both exploratory analysis and machine learning. Implementing these actions can reduce preventable attrition, improve employee engagement, and strengthen organizational performance through data-driven workforce management.

High-Risk Employee Profile

By combining the top four predictive drivers identified by the Random Forest model, we can define a clear employee profile most susceptible to attrition. This highlights the operational conditions under which turnover risk becomes significantly elevated.

- **Tenure:** 4–6 Years
- **Project Load:** 5+ Projects
- **Monthly Hours:** >175 Hours
- **Performance Evaluation:** High
- **Risk Level:** Elevated

Profile Insight: High-performing employees with sustained heavy workloads and mid-tenure service represent the most significant attrition risk segment. Turnover in this group appears closely associated with workload pressure, burnout risk, and career plateau challenges, making them a critical target for proactive retention initiatives.

Expected Business Outcomes

Reduced Burnout: Lower burnout-related attrition through sustainable workload management.

Targeted Retention: Improved retention of high-performing and mid-tenure (4–6 year) employees.

Workforce Stability: Higher overall employee engagement, productivity, and team stability.

Proactive HR: Earlier identification of flight risks through predictive Random Forest monitoring.

Cost Savings: Significantly lower recruitment, onboarding, and replacement costs.

Strategic Outcome: By addressing the root causes of attrition, Salifort Motors can build a more engaged, productive, and resilient workforce.